

IM E K FEDOROVA, Russian Federation

WMO: 209460

Lat: 70.447N

Lon: 59.091E

Elev: 12

StdP: 101.18

Time Zone: 3.00 (E03)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-28.6	-34.5	0.1	-30.9	-31.9	0.2	-28.7	19.6	-10.8	17.8	-9.0	5.1	130	1.101

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.9	15.8	13.2	13.2	11.5	11.2	9.9	13.6	15.9	11.6	13.2	10.0	11.1	7.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	9.0	14.5	10.5	7.9	12.4	9.2	7.2	10.9	38.3	15.9	33.2	13.2	29.4	11.2	19.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.8	15.1	13.7	DB	-30.5	18.6	4.8	3.8	-34.0	21.3	-36.8	23.5	-39.5	25.6	-43.0	28.4	(4)
			WB	-30.8	15.1	4.6	2.4	-34.1	16.8	-36.7	18.2	-39.3	19.6	-42.6	21.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.3	-15.6	-14.5	-13.6	-8.3	-3.8	2.1	6.7	6.8	4.8	-0.7	-6.6	-9.7
	DBStd	9.46	7.49	7.75	7.80	5.37	3.68	3.17	3.83	2.39	2.16	3.34	5.22	6.09
	HDD10.0	5248	793	685	731	549	427	239	120	105	157	333	499	611
	HDD18.3	8266	1051	918	989	799	685	487	361	359	407	591	749	869
	CDD10.0	25	0	0	0	0	0	2	17	5	1	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.1	7.1	7.6	7.3	7.0	6.6	6.0	5.8	6.5	6.8	7.8	8.4	8.3
Precipitation	PrecAvg	193	8	11	8	8	11	16	20	25	27	31	20	16
	PrecMax	332	25	31	27	26	38	44	49	76	72	59	45	42
	PrecMin	62	0	0	0	0	0	1	0	2	6	10	4	4
	PrecStd	72	7	10	8	6	8	13	17	17	17	13	10	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.0	-0.1	0.2	2.6	6.3	17.3	20.9	15.9	11.5	5.7	1.7	0.8
		MCWB	-0.5	-0.7	-0.4	1.1	4.9	13.0	15.3	13.2	9.9	4.8	0.9	0.5
	2%	DB	-1.2	-1.4	-0.6	0.7	2.9	11.4	17.3	13.4	10.0	4.7	0.9	-0.3
		MCWB	-1.5	-1.7	-1.1	0.0	2.0	9.2	14.4	11.8	8.9	3.8	0.4	-0.6
	5%	DB	-2.6	-2.7	-1.4	0.1	1.5	8.0	14.9	11.7	8.7	4.2	0.2	-1.0
		MCWB	-2.9	-3.0	-1.7	-0.3	0.8	6.4	13.1	10.4	7.9	3.4	-0.4	-1.3
	10%	DB	-4.7	-3.9	-2.5	-0.7	0.7	6.1	12.5	10.1	7.8	3.6	-0.5	-1.9
		MCWB	-5.1	-4.2	-2.9	-1.0	0.2	5.1	11.1	9.2	6.9	2.7	-1.0	-2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	-0.4	-0.1	1.2	4.9	13.2	16.5	13.5	10.4	5.2	1.2	0.6
		MCDB	0.0	0.0	0.2	1.9	6.2	17.2	18.8	14.9	11.1	5.4	1.5	0.6
	2%	WB	-1.4	-1.6	-0.9	0.3	2.1	9.4	14.8	12.0	9.1	4.1	0.5	-0.4
		MCDB	-1.2	-1.4	-0.7	0.7	2.8	11.3	16.7	13.2	9.6	4.5	0.8	-0.3
	5%	WB	-2.8	-2.9	-1.7	-0.2	0.9	6.7	12.9	10.5	8.0	3.6	-0.3	-1.3
		MCDB	-2.6	-2.7	-1.4	0.2	1.2	7.9	15.0	11.6	8.5	4.1	0.1	-1.1
	10%	WB	-5.1	-4.2	-2.9	-1.0	0.3	5.1	11.1	9.3	7.1	2.8	-0.9	-2.3
		MCDB	-4.7	-3.9	-2.5	-0.7	0.6	6.1	12.6	10.0	7.7	3.4	-0.5	-2.0
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.3	6.0	6.1	4.1	3.3	3.8	2.9	2.3	2.7	4.1	5.5
		MCDBR	6.5	8.1	5.5	5.4	4.3	7.8	8.4	5.8	3.9	2.6	3.7	5.0
		MCWBR	6.6	8.2	5.5	5.2	3.9	5.8	5.6	4.7	3.3	2.6	3.8	5.1
	5% WB	MCDBR	6.3	8.1	5.1	5.1	3.9	7.9	7.9	5.6	3.5	2.6	3.7	5.0
		MCWBR	6.4	8.2	5.2	4.9	3.7	6.1	5.5	4.7	3.2	2.7	3.9	5.0
Clear-Sky Solar Irradiance	taub	N/A	0.242	0.291	0.320	0.336	0.336	0.371	0.352	0.316	0.261	N/A	N/A	
	taud	N/A	1.941	2.019	2.044	2.096	2.279	2.372	2.450	2.456	2.155	N/A	N/A	
	Ebn at Noon	0	423	677	791	831	850	801	774	695	416	0	0	
	Edh at Noon	0	42	84	115	125	108	96	79	58	33	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.27	1.73	3.83	5.47	5.67	4.92	3.05	1.49	0.37	0.00	0.00	
	RadStd	0.00	0.05	0.20	0.32	0.52	0.56	0.61	0.29	0.11	0.05	0.00	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47)	Regional (2 neighbors)	+1.72	+2.78	+3.36	+1.64	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page